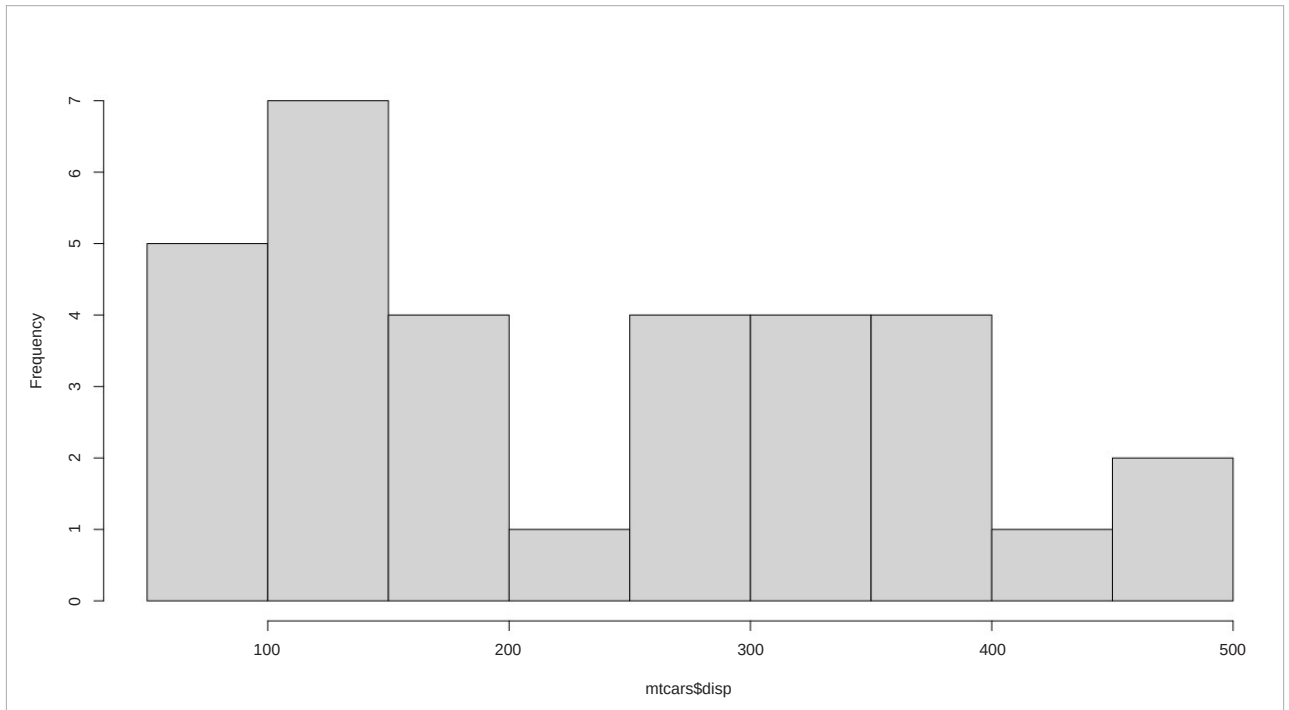


Figure 1: *mtcars\$disp* histogram

$f(x) = 1/b-a$

...where $a \leq x \leq b$. The *runif()* function generates random deviates, while the *dunif()* function gives the density, as illustrated below:

```
> d <- runif(10, min=0, max=1)
> d
[1] 0.48730291 0.26676446 0.15025110 0.77975962 0.42828890
0.05255129
[7] 0.54070173 0.54715839 0.42703064 0.31628728

> dunif(d)
[1] 1 1 1 1 1 1 1 1 1 1
```

These functions accept the following common arguments:

Arguments	Description
x, q	A vector of quantiles
p	A vector of probabilities
n	Number of observations
log, log.p	A TRUE logical value indicates probabilities as <i>log(p)</i>
lower.tail	A TRUE logical value that probabilities are $P[X \leq x]$

The specific set of arguments for the uniform distribution is given below:

Arguments	Description
min, max	Finite lower and upper distribution limits

Binomial

The density of the binomial distribution with probability 'p' and size 'n' is given by:

$$p(x) = \text{select}(n, x) p^x (1-p)^{(n-x)}$$

...where 'x' varies from 0 to n. The $p(x)$ is calculated using the Loader's algorithm. The *pbinom()* function gives the distribution function while the *rbinom()* function provides random values, as shown below:

```
> rbinom(5, 5, 1)
[1] 5 5 5 5 5

> rbinom(5, 5, 0.5)
[1] 3 2 2 4 2

> rbinom(5, 5, 0.25)
[1] 0 3 0 2 2
```

An example of *pbinom()* function for five successes in 10